AWS Outposts	Amazon EKS Anywhere
Works on AWS designed infrastructure only	Works on VMWare or custom infrastructure (vm or bare metal)
Costly due to usage of AWS hardware	Cheaper as we can reuse existing custom infrastructure
Fully connected network connectivity for hybrid cloud	Fully connected, partially-connected, fully disconnected network connectivity for hybrid cloud
Extends AWS services to data center	Flexible to extend EKS in on-premises data centers
Flexible to integrate with AWS services in hybrid cloud	Need to customize to suit custom infrastructure

Table 1: A comparison of AWS Outposts and Amazon AKS Anywhere

AWS Outposts can be used to deploy EKS instances on AWS infrastructure, using the same architecture — control plane, data plane and worker nodes or instances. AWS EKS Anywhere can easily migrate existing Kubernetes services running on-premise (like VMWare Tanzu Kubernetes Grid) to run on an AWS EKS distribution.

AWS EKS Anywhere can be easily considered as a direct competitor of Google Anthos or Azure Arc. It runs self-managed services on VMWare vSphere and is deployable on any cloud platform. It facilitates application service communication through proxying, using App Mesh and Kubernetes distribution and deployment services with Flux services.

Spinnaker for continuous delivery

Spinnaker is a popular continuous deployment/delivery framework suitable for multi-cloud environments. It supports build pipelines with Jenkins, Travis CI, Docker or Git event based triggers. Conceived by Netflix, Spinnaker is now part of the Netflix open source software (OSS) service.

For Kubernetes deployment on the AWS platform, Spinnaker is flexible and handy for quicker integration with Jenkins for continuous integration and deployment to multi-stage environments. The solution blueprint involves developing code in a Cloud9 workspace, and checking into GitHub to trigger a code build using Jenkins, which can create a container image using the CloudFormation template or build script. This gets pushed to the Elastic Container Registry (ECR), from where the Spinnaker pipeline triggers builds when a new Docker image gets added to ECR.

The Spinnaker pipeline involves configuring Kubernetes deployment, generating Kubernetes deployment (a.k.a. Spinnaker bake) for different stages like dev, test and prod, and then deploying the image to Kubernetes pod and to the development environment. After validation checks, manual approval (or judgement) is required to promote the image to the next stage (test and then to production). This ensures that the application container is not promoted to the next stage without validation or approval.

Spinnaker uses Helm v2 for the managing and baking activities of Kubernetes deployments. It has a Helm template for application deployment and a YAML script for dev, test and prod activities. By integrating Spinnaker with Jenkins, we can easily create a container deployment pipeline with CI/CD as a toolchain process for faster time to market. END

References

- *https://aws.amazon.com/blogs/opensource/continuous-delivery-spinnaker-amazon-eks/*
- *https://vticloud.io/en/aws-phan-hoi-anthos-va-azure-arc-voi-amazon-eks-anywhere/*
- *https://aws.amazon.com/eks/pricing/*
- *https://learn.microsoft.com/en-us/training/modules/intro-to-kubernetes/*
- *https://aws.github.io/aws-eks-best-practices/cluster-autoscaling/*
- *https://docs.aws.amazon.com/eks/latest/userguide/autoscaling.htm*

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.